

Final CFG

Below is the completed CFG in BNF with all EBNF removed, left recursion removed, and collisions removed. Additionally, identifiers have been replaced with shorthand version.

P	program I ; var B begin G end
I	LX
X	LX
X	DX
X	λ
B	C : K ;
C	IZ
Z	, C
Z	λ
K	integer
G	HV
V	G
V	λ
H	W
H	A
W	print(JI);
J	"value=",
J	λ
A	I = E ;
E	TQ
Q	+TQ
Q	-TQ
Q	λ
T	FR
R	*FR
R	/FR
R	λ
F	I
F	N
F	(E)
N	SDY
Y	DY
Y	λ
S	+
S	-
S	λ
D	0
D	1
D	2

D	3
D	4
D	5
D	6
D	7
D	8
D	9
L	a
L	b
L	c
L	d
L	l
L	f

Members of FIRST and FOLLOW

Below are the members of FIRST and FOLLOW of all non-terminals. In addition, the shorthand versions of the non-terminals are paired with their original names to see which is which.

Original	Symbol	FIRST	FOLLOW
<prog>	P	program	\$
<identifier>	I	a b c d l f	; , :) = * / + -
<X>	X	λ 0 1 2 3 4 5 6 7 8 9 a b c d l f	; , :) = * / + -
<dec-list>	B	a b c d l f	begin
<dec>	C	a b c d l f	:
<Z>	Z	λ ,	:
<type>	K	integer	;
<stat-list>	G	print a b c d l f	end
<V>	V	λ print a b c d l f	end
<stat>	H	print a b c d l f	print a b c d l f end
<write>	W	print	print a b c d l f end
<str>	J	λ "value=",	a b c d l f
<assign>	A	a b c d l f	print a b c d l f end
<expr>	E	(+ - 0 1 2 3 4 5 6 7 8 9 a b c d l f	;))
<expr>'	Q	+ - λ	;))
<term>	T	(+ - 0 1 2 3 4 5 6 7 8 9 a b c d l f	+ - ;)
<term>'	R	* / λ	+ - ;)
<factor>	F	(+ - 0 1 2 3 4 5 6 7 8 9 a b c d l f	* / + - ;)
<number>	N	+ - 0 1 2 3 4 5 6 7 8 9	* / + - ;)
<Y>	Y	λ 0 1 2 3 4 5 6 7 8 9	* / + - ;)
<sign>	S	+ - λ	0 1 2 3 4 5 6 7 8 9
<digit>	D	0 1 2 3 4 5 6 7 8 9	0 1 2 3 4 5 6 7 8 9 a b c d l f * / + - ;) , : =
<letter>	L	a b c d l f	0 1 2 3 4 5 6 7 8 9 a b c d l f * / + - ;) , : =

Predictive Parsing Table

	a	b	c	d	l	f
P						
I	LX	LX	LX	LX	LX	LX
X	LX	LX	LX	LX	LX	LX
B	C:K;	C:K;	C:K;	C:K;	C:K;	C:K;
C	IZ	IZ	IZ	IZ	IZ	IZ
Z						
K						
G	HV	HV	HV	HV	HV	HV
V	G	G	G	G	G	G
H	A	A	A	A	A	A
W						
J	λ	λ	λ	λ	λ	λ
A	I=E;	I=E;	I=E;	I=E;	I=E;	I=E;
E	TQ	TQ	TQ	TQ	TQ	TQ
Q						
T	FR	FR	FR	FR	FR	FR
R						
F	I	I	I	I	I	I
N						
Y						
S						
D						
L	a	b	c	d	l	f

	0	1	2	3	4	5	6	7	8	9
P										
I										
X	DX	DX	DX	DX	DX	DX	DX	DX	DX	DX
B										
C										
Z										
K										
G										
V										
H										
W										
J										
A										
E	TQ	TQ	TQ	TQ	TQ	TQ	TQ	TQ	TQ	TQ
Q										
T	FR	FR	FR	FR	FR	FR	FR	FR	FR	FR
R										
F	N	N	N	N	N	N	N	N	N	N
N	SDY	SDY	SDY	SDY	SDY	SDY	SDY	SDY	SDY	SDY
Y	DY	DY	DY	DY	DY	DY	DY	DY	DY	DY
S	λ	λ	λ	λ	λ	λ	λ	λ	λ	λ
D	0	1	2	3	4	5	6	7	8	9
L										

	+	-	*	/	;	:	(	)	,	=
P										
I										
X	λ	λ	λ	λ	λ	λ		λ	λ	λ
B										
C										
Z						λ			,C	
K										
G										
V										
H										
W										
J										
A										
E	TQ	TQ					TQ			
Q	+TQ	-TQ			λ			λ		
T	FR	FR					FR			
R	λ	λ	*FR	/FR	λ			λ		
F	N	N					(E)			
N	SDY	SDY								
Y	λ	λ	λ	λ	λ			λ		
S	+	-								
D										
L										

	"value=",	print	integer	program	begin	end	\$
P				program I ; var B begin G end			
I							
X							
B							
C							
Z							
K			integer				
G		H V					
V		G				λ	
H		W					
W		print (J I) ;					
J	"value=",						
A							
E							
Q							
T							
R							
F							
N							
Y							
S							
D							
L							